GH

Principle of Mathematical Induction

Prove that

$P(n) : 1+2+3+\ldots\ldots n = \frac{n(n+1)}{2}$ for all $n \geq 1$

1) Base case prove $P(1)$ is true
2) Induction step
   a) Assume that the theorem is true for some value k i.e $P(k)$ is true
   b) Prove that $P(k+1)$ is true

Step 1 $P(1)$ is true

$LHS = n$

$n = 1$

$RHS = \frac{1(1+1)}{2} = \frac{2}{2} = 1$

$LHS = RHS$ —

Step 2 a) Assume for some k $P(k)$

$1+2+3+4+\ldots\ldots k = \frac{(k)(k+1)}{2}$

assume that this is true

b) Prove for $P(k+1)$

what we want to prove

$1+2+3+\ldots\ldots+k+(k+1) = \frac{(k+1)(k+1+1)}{2}$

$LHS = \underbrace{1+2+3\ldots\ldots k}_{\frac{k(k+1)}{2}} + (k+1)$

$= \frac{k(k+1)+2(k+1)}{2}$

$LHS = \frac{(k+2)(k+1)}{2}$ $RHS = \frac{(k+1)(k+2)}{2}$

Hence Proved $LHS = RHS$

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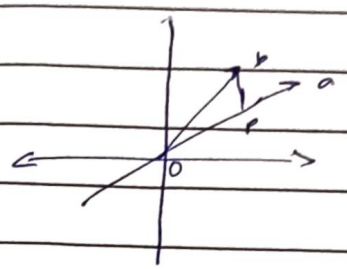
Hyper plane -> Is an affine subspace of dimension $(d-1)$

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots \beta_d x_d = 0$$



$e = b - p$

$p = x a$

e perpendicular to a

$a^T e = 0$

$a^T (b - xa) = 0$

$a^T b - a^T x a = 0$

$a^T b = a^T x a$

$x = \dfrac{a^T b}{a^T a}$

$p = a x$

$p = \dfrac{a a^T b}{a^T a}$

$e = b - p$

$p = x a$

$a^T e = 0$

$e^T (b - xa) = 0$

$p = PM\, b$

$PM = \dfrac{a a^T}{a^T a}$

$a^T a = [1, 2] \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ => Scalar

$a a^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [1 \quad 2]$ => vector

$PM = A (A^T A)^{-1} A^T$